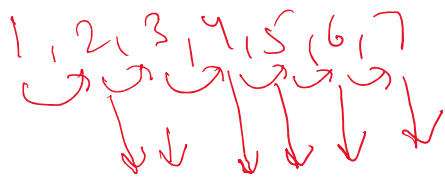


Store last element, $= 7$, shift all to the right



$$arr[j] = arr[j-1];$$

1, 1, 2, 3, 4, 5, 6

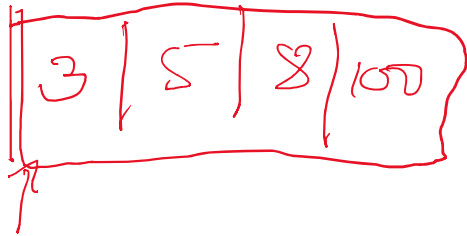
↓ $arr[0] = \text{last element}$

7, 1, 2, 3, 4, 5, 6

Basically, I have done the rotation for one time,
but I need to rotate this for k times.

↳ 3

✓ Max / Min



→ `int maxi = arr[0];`

// I will store the maxi variable with some values

→ Iterate through the whole array;

→ `for (int i = 1; i < n; i++) {
 if (arr[i] > maxi) {
 maxi = arr[i];
 }`

→ Rotate the array by k steps →

→ [1, 2, 3, 4, 5, 6, 7]

→ (k = 3)

Original
1 2 3 4 5 6 7

Step 1
7 1 2 3 4 5 6

Step 2
6 7 1 2 3 4 5

Step 3
5 6 7 1 2 3 4

Idea is that rotate array by 1 steps

→ Repeat for k times

→ 7, 1, 2, 3, 4, 5, 6 → 4, 5, 6, 7, 1, 2, 3
→ 6, 7, 1, 2, 3, 4, 5 → 3, 4, 5, 6, 7, 1, 2
→ 5, 6, 7, 1, 2, 3, 4 → 2, 3, 4, 5, 6, 7, 1
→ 1, 2, 3, 4, 5, 6, 7

Steps

→ Store last element

→ last element = 7

Shift all elements to the right

7, 1, 2, 3, 4, 5, 6

(last Element = 6)

[7, 1, 2, 3, 4, 5]

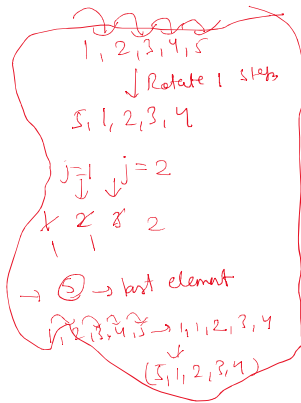
↓ change first element as last element

[6, 1, 2, 3, 4, 5]

next step do the same but last element become 5 here
and you get a array that you need

```
public void rotate(int[] nums, int k)
```

```
→ int n = nums.length;
for (int i = 0; i < k; i++) {
    int last = nums[n-1];
    for (j = n-1; j > 0; j--) {
        nums[j] = nums[j-1];
    }
    nums[0] = last;
}
```



```
int arr = {1, 2, 3, 4, 5};
```

```
int n = arr.length;
```

```
int last = arr[n-1];
```

```
for (i = n-1; i > 0; i--) {
```

```
    arr[i] = arr[i-1];
}
arr[0] = last;
```

5 = 4
4 = 3
3 = 2
2 = 1

```
for (int i = 0; i < k; i++) {
```

⋮

Approach II :-

(size of array = 5)

(Taking extra array)

→ new array temp

→ temp size is equal arr

→ Put last element at temp[0];

→ copy remaining element.

nums = [1, 2, 3, 4, 5] temp = [-1, -1, -1, -1]

temp = [5, +1, -1, -1, -1]

S.S.: temp[1] = nums[0]; temp $\Rightarrow$
temp[2] = nums[1] [5, 1, 2, 3, 4]
temp[3] = nums[2]
temp[4] = nums[3]

int [] arr = {1, 2, 3, 4, 5}

int n = arr.length;

int [] temp = new int[n]; temp = [-1, -1, -1, -1]

temp[0] = arr[n-1] $\rightarrow$ [5, -1, -1, -1, -1]

```
for(int i=0; i<n; i++){  
    temp[i+1] = arr[i];  
}
```

H.W.

With approach I, try to do the dry run for k=12, and length of array is only 5 itself. [5, 4, 1, 2, 3]

- I will rotate by k steps
- for each individual steps I will shift all elements right.
- Put last element at index 0;